INCH-POUND

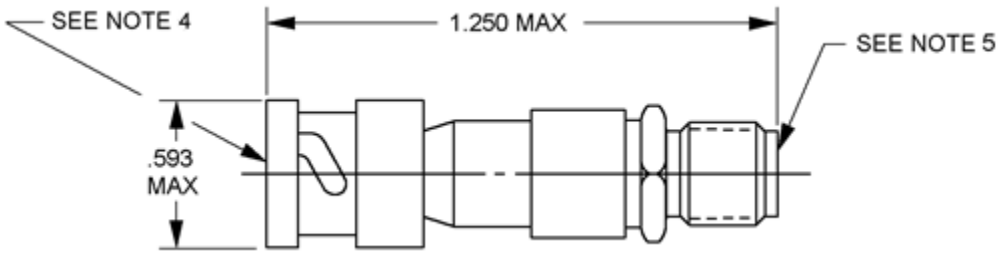
MIL-PRF-55339/44C
w/AMENDMENT 1
03 September 2024
SUPERSEDING
MIL-PRF-55339/44C
28 January 2015

PERFORMANCE SPECIFICATION

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, IN-LINE, (BETWEEN SERIES SMA JACK TO SERIES BNC PLUG), CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55339.



NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial representations are for reference purposes only.
3. BNC pin contact interface, in accordance with MIL-STD-348.
4. SMA socket contact, in accordance with MIL-STD-348.

FIGURE 1. General configuration.

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ENGINEERING DATA:

Design and construction:

General configuration: See figure 1.

Impedance: 50 ohms, nominal.

Working voltage:

Sea level: 335 V rms.
70,000 feet: 85 V rms.

Frequency range: 0 to 4 GHz.

Temperature range: -65°C to +165°C.

REQUIREMENTS (installation torque of 4 to 6 inch-pounds).

Series BNC coupling nut shall be nickel plated in accordance with SAE-AMS-QQ-N-290, 200-300 microinches, thickness.

Dimensions: See figure 1 and MIL-STD-348.

Center contact retention:	Series BNC	Series SMA
Axial force: (pounds, minimum):	6	6
Torque: (inch-ounces, minimum):	Not applicable	Not applicable

Force to engage and disengage:	Series BNC	Series SMA
Longitudinal force (pounds, maximum):	3	Not applicable
Torque: (inch-pounds, maximum):	2.5	2

Mating characteristics:

Series BNC:

Outer contact:

Minimum test pin inner diameter: .319 inch, maximum.
Pin finish: 16 microinches.
Insertion force: 5 pounds, maximum.
Insertion depth: .093 inch, minimum.
Number of insertions: 1.

Contacts with slotted members:
Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Series SMA:

Center contact (socket):
Oversize test pin diameter: .0375 + .0001.
Insertion depth: .030/.045 inch, minimum.
Number of insertions: 3.
Pin finish: 16 microinches.

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Maximum Test pin (insertion force test):
Steel test pin diameter: .0370 + .0001.
Pin finish: 16 microinches.
Insertion force: 3 pounds, maximum.
Number of insertions: 3.

Minimum test pin (withdrawal force):
Steel test pin diameter: .0355 - .0001.
Pin finish: 16.
Withdrawal force: 1 ounce, minimum.
Number of withdrawals: 1.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Permeability: <2.0.

Seal:
Pressurized: Not applicable.
Weatherproof: Not applicable.

Insulation resistance: 5,000 megohms, minimum.

VSWR: 1.30:1 maximum at .5 to 4 GHz.

RF leakage (total): -55 dB, minimum, 2 to 3 GHz.

RF insertion loss: .2 dB, maximum, 3 GHz (.115 $\sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz).

Durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Dielectric withstanding: Test voltage: 1,500 V rms, minimum (sea level).

Contact resistance (milliohms, maximum):

Contract	Initial	After
Center	4.1 1/	6.0
Outer	2.2	Not applicable

1/ Two center contacts must be mated to the center conductor under test, therefore doubling “center contact” resistance.

Vibration, high frequency: MIL-STD-202-204, test condition D, interruptions – 1 µs, maximum.

Shock (specified pulse): MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition C.

Moisture resistance: 200 megohms, minimum.

Corona level:
Voltage: 250 V, minimum.
Altitude: 70,000 feet, minimum.

RF high potential withstanding voltage:
RF voltage: 670 V rms, minimum.

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Frequency: 5 MHz, minimum.

Salt spray (corrosion): Test condition B.

Coupling mechanism retention force:

Series BNC	Series SMA
100 pounds, minimum	Not applicable

Group qualification: See table I.

TABLE I. Group qualifications.

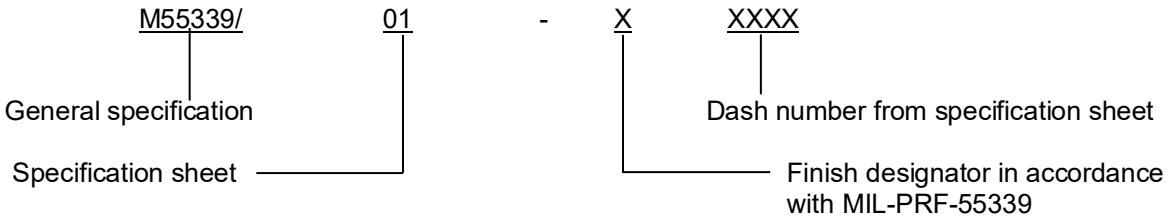
Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	M55339/44-30001 M55339/44-50001	M55339/44-30001 M55339/44-50001

Marking: As specified in MIL-PRF-55339.

Part or Identifying Number (PIN):

M55339/44-30001.
M55339/44-50001.

PIN configuration:



Referenced documents: In addition to MIL-DTL-55339, this document references the following:

MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-348

SAE-AMS-QQ-N-290

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2024-196)

Review activities:
Army - AR, AT
Navy - AS, MC
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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